

PCB ASSEMBLY INSTRUCTIONS

Project : PelletPirate ESP32-S3 Breadboard (ESP_Breadboard)

Revision: esp32-v2

Date : 2026-06-29

Service : Turnkey PCB Fabrication + Assembly (PCBWay sourcing components)

1. BOARD SPECIFICATIONS

Layers 2 (copper top + bottom)
Board size 30.0 mm x 70.0 mm
Thickness 1.6 mm
Material FR-4
Copper weight 1 oz
Surface finish ENIG (REQUESTED - see note in Section 6)
Min track/space ~0.20 mm / 0.15 mm (standard)
Min hole 0.30 mm
Drill files PTH and NPTH provided SEPARATELY.
The 2x non-plated 0.65 mm holes are USB-C locating
pegs and MUST remain non-plated (see NPTH drill file).

2. ASSEMBLY SCOPE (READ FIRST - MIXED TECHNOLOGY)

This board is MIXED-TECHNOLOGY. Quote/assemble BOTH processes:

A) SMT (reflow), TOP side:

ESP1, U1, J8 (signal pads), LED1,
C17, C18, C37, C38, C39, C40,
R30, R31, R35, R36, R37, R38, R39

B) THROUGH-HOLE (hand or selective solder) - DO NOT SKIP:

J1, J2 ... 20-pin pin headers, mounted on the BOTTOM side
S1, S2 ... Omron B3F tactile switches
J8 ... USB-C is HYBRID: SMD signal pads (top) PLUS through-hole
shield/ground posts. Solder the THT shield posts.

NOTE: A default "SMT only" quote will MISS the through-hole parts.
Please include through-hole soldering in the quote.

3. SIDE ASSIGNMENT

TOP all SMD components (see 2A)
BOTTOM .. J1 and J2 pin headers (pins protrude downward for breadboard use)
>> Do NOT flip J1/J2 to the top side. <<

4. CRITICAL PARTS - NO SUBSTITUTION

Do NOT substitute the following. If unavailable, CONTACT US before building.

Ref	MPN	Mfr	Reason
ESP1	ESP32-S3-WROOM-1U-N16R8	Espressif	Exact variant required: 16 MB flash, 8 MB OCTAL PSRAM, U.FL antenna (-1U). DO NOT substitute with N8R2, N16R16V (1.8V PSRAM), or WROOM-1 (PCB antenna).
J8	USB4105-GF-A	GCT	Footprint-specific USB-C.

U1	USBLC6-2SC6	STMicro	Exact ESD device.
C37	C0805C105K3RACTU (1uF)	KEMET	25V X7R minimum.
C39	C0805C105K3RACTU (1uF)	KEMET	25V X7R minimum.
C38	C1206C106K3RACTU (10uF)	KEMET	25V X7R minimum.
C40	C0805C475M3RACTU (4.7uF)	KEMET	25V X7R minimum.
C17	C0402C104K3RACTU (0.1uF)	KEMET	25V X7R.
C18	C0402C104K3RACTU (0.1uF)	KEMET	25V X7R.

CAPACITOR RULE: All capacitors must be 25V-rated, X7R dielectric (or better).
Do NOT substitute Y5V / Z5U dielectrics at any value.

5. SUBSTITUTION ALLOWED (equivalent parts OK)

Resistors (thick/thin film, 1% tolerance, SAME 0402 package):

R30, R31, R35, R36 ... 10 kohm

R37, R38 5.1 kohm

R39 1 kohm

LED1 ... Green LED, 0603, same footprint/polarity (LTST-C191KGKT or equiv.)

Switches S1, S2 ... Omron B3F-1000 or mechanically/electrically equivalent.

6. ORIENTATION / POLARITY - VERIFY THESE

Confirm pin-1 / polarity / keying for:

ESP1 module pin 1 orientation

U1 USBLC6-2SC6 pin 1 (SOT-23-6)

LED1 cathode / polarity

J8 USB-C connector orientation / keying

Refer to the centroid/pick-and-place file and (if provided) the assembly drawing for placement and rotation.

7. SPECIAL PROCESS NOTES

- ESP1 has an exposed thermal pad (EP / pad 41) on its underside, with a via array to the bottom ground plane. Apply correct paste volume; ensure good EP reflow without bridging to adjacent perimeter pads.
- USB-C (J8) signal pads are 0.5 mm pitch (fine pitch). ENIG finish is requested to improve coplanarity and assembly yield.
- There are NO do-not-populate (DNP) parts. Populate all references.

8. FILES PROVIDED

- Gerbers (RS-274X): F.Cu, B.Cu, F/B Mask, F/B Silk, F/B Paste, Edge.Cuts
- Drill: *-PTH.drl and *-NPTH.drl (separate)
- Drill maps: *-drl_map.pdf
- Centroid / pick-and-place: *-centroid.csv
- Assembly BOM: (provided separately)

9. DFM / QUESTIONS

Please send any DFM findings or proposed part substitutions for written approval BEFORE starting assembly. Do not substitute Section 4 parts.

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END OF ASSEMBLY INSTRUCTIONS
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PelletPirate ESP32-S3 Breadboard - 3D Render

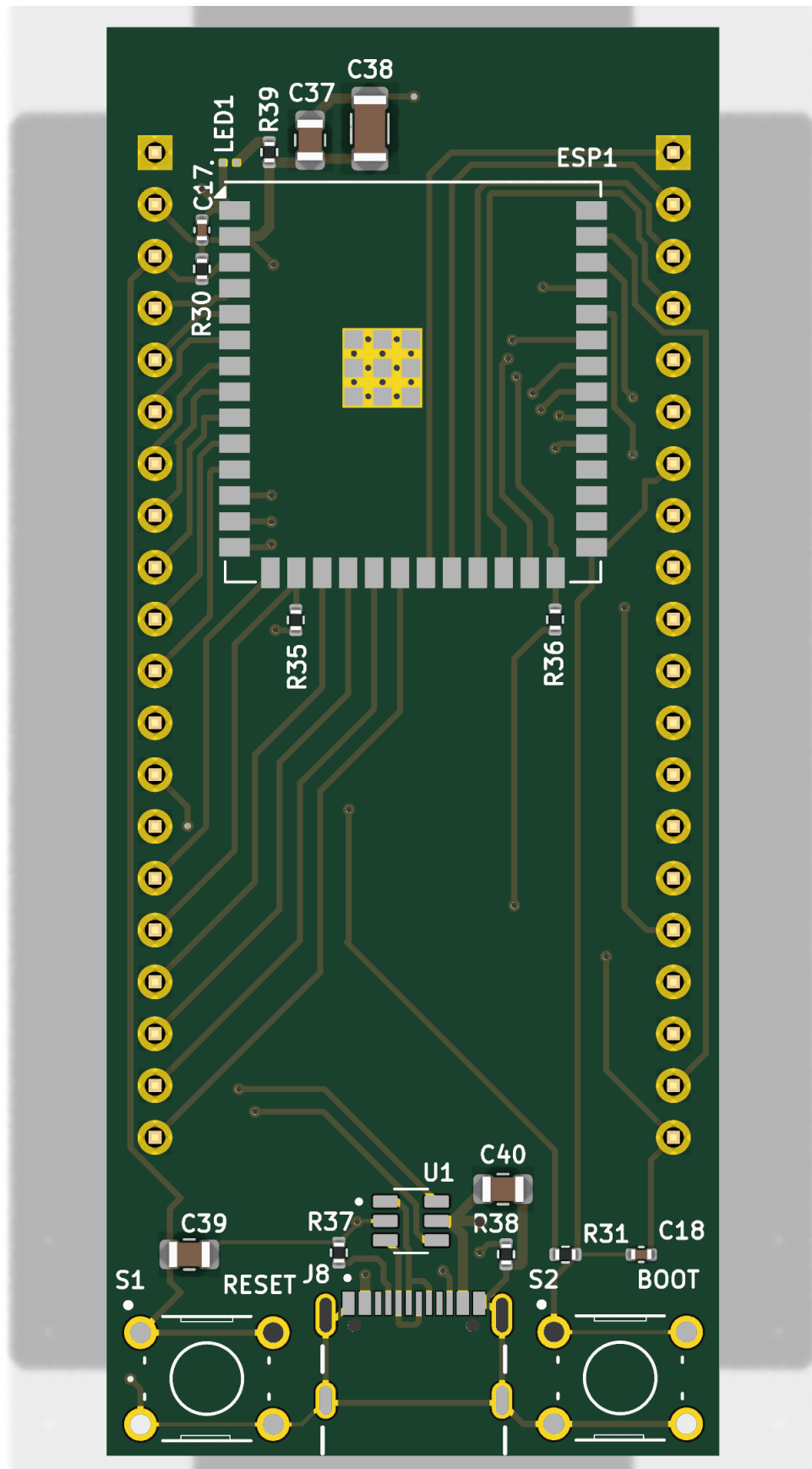


Figure 1 - Top view

PelletPirate ESP32-S3 Breadboard - 3D Render

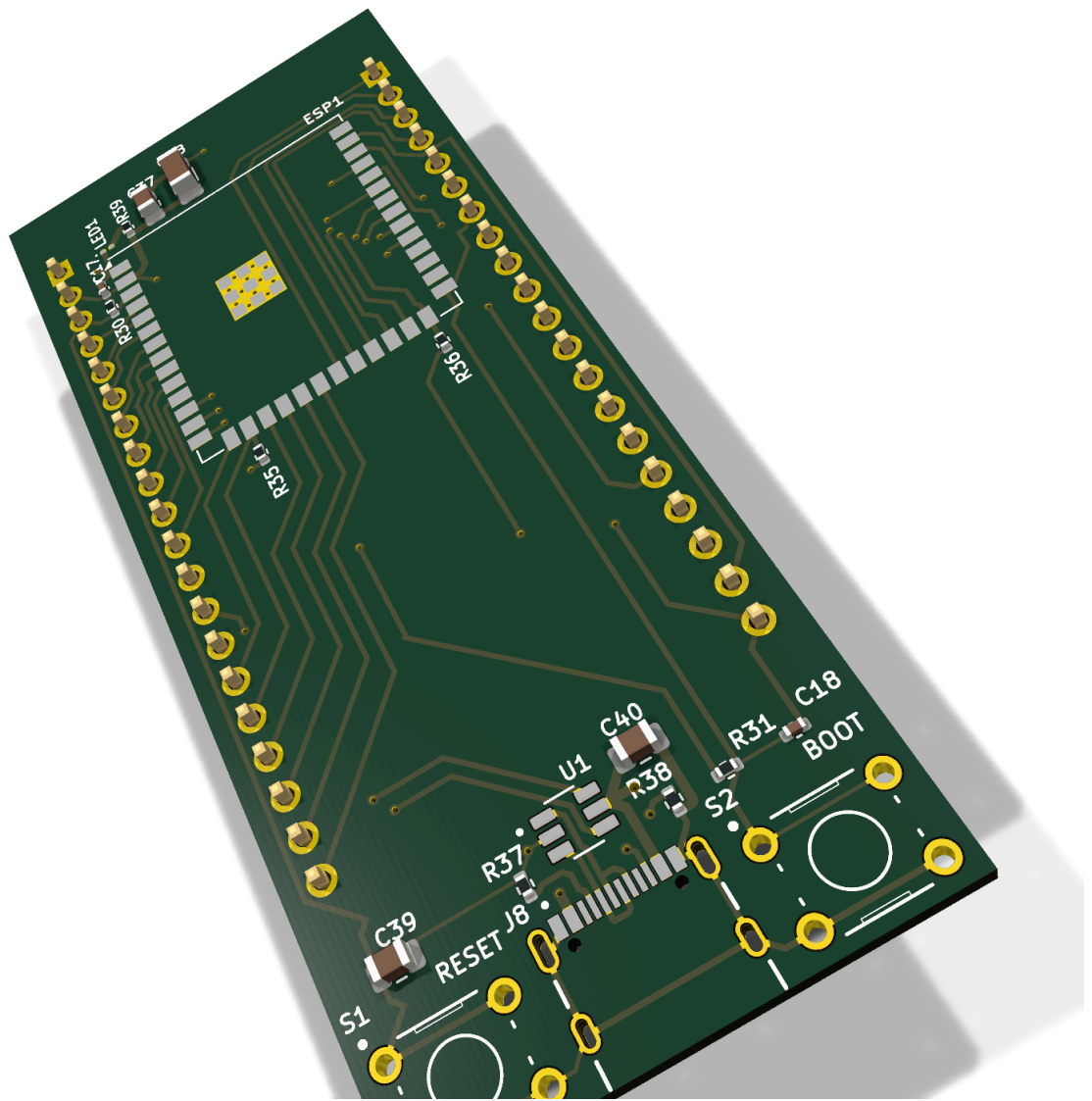


Figure 2 - Perspective view

PelletPirate ESP32-S3 Breadboard - 3D Render

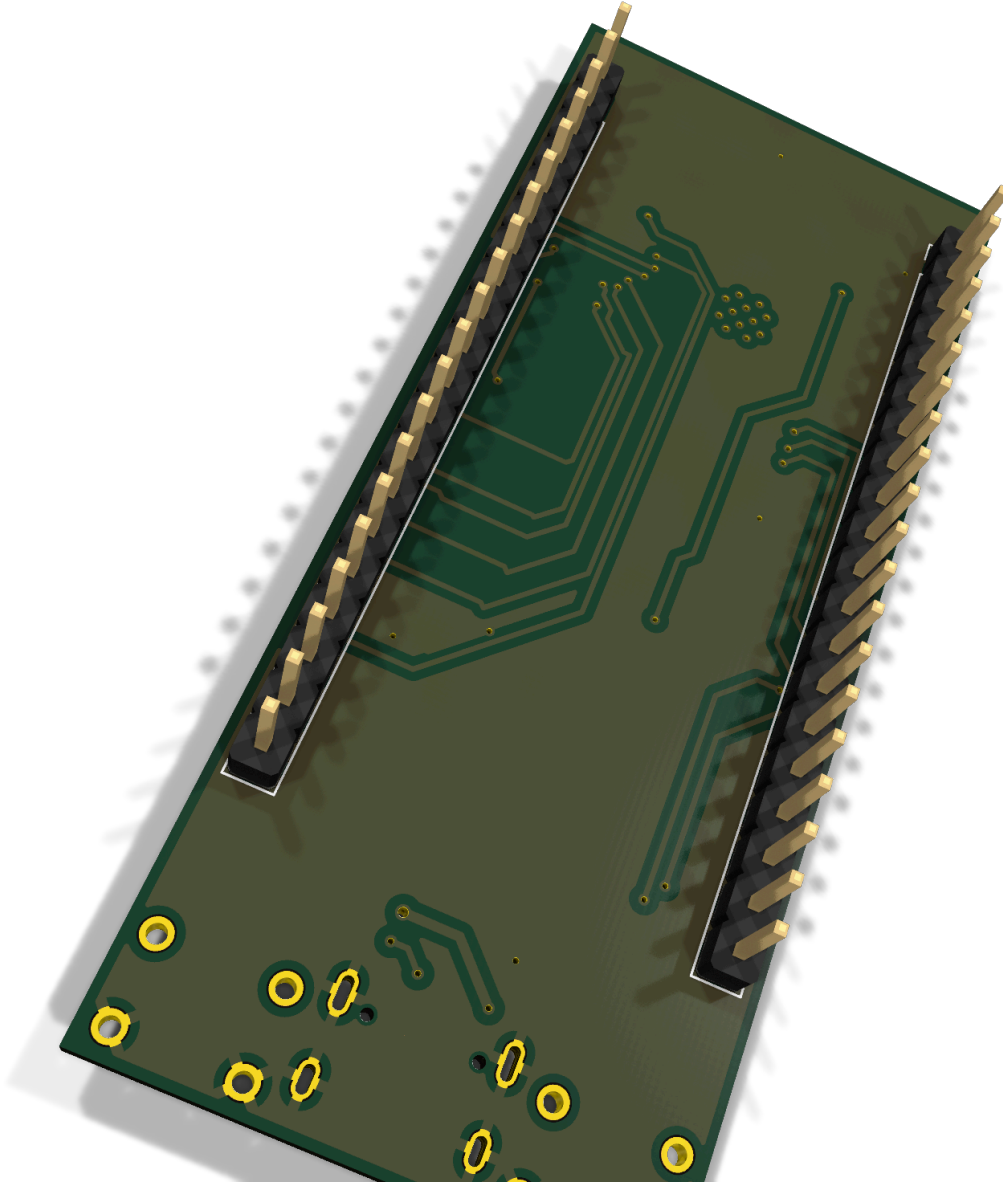


Figure 3 - Bottom view (breadboard pin rails on bottom side)